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Risk models 2

Syllabus objectives

- 1.2 Compound distributions and their application in risk modelling
 - 1.2.1 Construct models appropriate for short-term insurance contracts in terms of the numbers of claims and the amounts of individual claims.
 - 1.2.2 Describe the major simplifying assumptions underlying the models in 1.2.1.
 - 1.2.4 Derive the mean, variance and coefficient of skewness for compound binomial, compound Poisson and compound negative binomial random variables.
 - 1.2.5 Repeat 1.2.4 for both the insurer and the reinsurer after the operation of simple forms of proportional and excess of loss reinsurance.

0 Introduction

In this chapter we will look at some of the practical applications of risk models. We start by looking at how the models can be adapted for situations involving reinsurance. A section on the individual risk model is followed by some more complex examples of the use of risk models in practice.

1 Aggregate claim distributions under proportional and individual excess of loss reinsurance

In [Chapter 19](#), we introduced the notation S to denote the aggregate claim amount random variable, ie:

$$S = X_1 + X_2 + \cdots + X_N$$

where N denotes the number of claims and X_i denotes the amount of the i th claim.

Here we extend this concept to consider the situation when reinsurance is in force. We will use the following notation:

- Y_i is the amount paid by the insurer in respect of the i th claim
- Z_i is the amount paid by the reinsurer in respect of the i th claim
- $S_I = Y_1 + Y_2 + \cdots + Y_N$ is the aggregate claim amount paid by the insurer
- $S_R = Z_1 + Z_2 + \cdots + Z_N$ is the aggregate claim amount paid by the reinsurer.

The formulae that we derived for the mean, variance and MGF of S can be adapted to cover the reinsurance situation by replacing X by Y or Z , as appropriate. For example:

$$E(S_I) = E(N)E(Y)$$

$$\text{var}(S_I) = E(N)\text{var}(Y) + \text{var}(N)[E(Y)]^2$$

$$M_{S_I}(t) = M_N(\log M_Y(t))$$

1.1 Proportional reinsurance

The distribution of the number of claims involving the reinsurer is the same as the distribution of the number of claims involving the insurer, as each pays a defined proportion of every claim.

For a retention level α ($0 \leq \alpha \leq 1$), the i th individual claim amount for the insurer is αX_i and for the reinsurer is $(1 - \alpha)X_i$.

In other words:

$$Y_i = \alpha X_i$$

$$Z_i = (1 - \alpha)X_i$$

So:

$$\begin{aligned}
 S_I &= Y_1 + Y_2 + \cdots + Y_N \\
 &= \alpha X_1 + \alpha X_2 + \cdots + \alpha X_N \\
 &= \alpha(X_1 + X_2 + \cdots + X_N) \\
 &= \alpha S
 \end{aligned}$$

and:

$$\begin{aligned}
 S_R &= Z_1 + Z_2 + \cdots + Z_N \\
 &= (1 - \alpha)X_1 + (1 - \alpha)X_2 + \cdots + (1 - \alpha)X_N \\
 &= (1 - \alpha)(X_1 + X_2 + \cdots + X_N) \\
 &= (1 - \alpha)S
 \end{aligned}$$

ie the aggregate claims amounts are αS and $(1 - \alpha)S$ respectively.



Question

- (i) Show that under a proportional reinsurance arrangement where the direct writer retains a proportion α , the MGF of the net individual claim amount Y paid by the direct insurer is $M_X(\alpha t)$.
- (ii) Hence give a formula for $M_{S_I}(t)$ when the number of claims follows a Poisson distribution with mean 25 and individual claim amounts are exponentially distributed with mean 1,000.

Solution

- (i) **MGF of Y**

Under this arrangement, $Y = \alpha X$. So:

$$M_Y(t) = E(e^{tY}) = E(e^{t\alpha X}) = E[e^{(t\alpha)X}] = M_X(\alpha t)$$

- (ii) **MGF of insurer's aggregate claim amount**

Since $N \sim \text{Poisson}(25)$:

$$M_N(t) = e^{25(e^t - 1)}$$

and the MGF of S_I is:

$$M_{S_I}(t) = M_N(\log M_Y(t)) = e^{25(\exp(\log M_Y(t)) - 1)} = e^{25(M_Y(t) - 1)}$$

In addition, since X is exponentially distributed with mean 1,000 (ie with parameter $\frac{1}{1,000}$):

$$M_X(t) = (1 - 1,000t)^{-1}$$

and hence:

$$M_Y(t) = M_X(\alpha t) = (1 - 1,000\alpha t)^{-1}$$

So the MGF of S_I is:

$$M_{S_I}(t) = e^{25[(1 - 1,000\alpha t)^{-1} - 1]}$$

1.2 Individual excess of loss reinsurance

The amount that an insurer pays on the i th claim under individual excess of loss reinsurance with retention level M is $Y_i = \min\{X_i, M\}$.

Equivalently:

$$Y = \begin{cases} X & \text{if } X \leq M \\ M & \text{if } X > M \end{cases}$$

The amount that the reinsurer pays is $Z_i = \max\{0, X_i - M\}$.

We can also write this as:

$$Z = \begin{cases} 0 & \text{if } X \leq M \\ X - M & \text{if } X > M \end{cases}$$

As previously stated, the insurer's aggregate claims net of reinsurance can be represented as:

$$S_I = Y_1 + Y_2 + \dots + Y_N$$

and the reinsurer's aggregate claims as:

$$S_R = Z_1 + Z_2 + \dots + Z_N \tag{20.1}$$

If, for example, $N \sim \text{Poi}(\lambda)$, S_I has a compound Poisson distribution with Poisson parameter λ , and the i th individual claim amount is Y_i . Similarly, S_R has a compound Poisson distribution with Poisson parameter λ , and the i th individual claim amount is Z_i .

Hence, for this compound Poisson distribution with reinsurance, we have:

$$E(S_I) = \lambda E(Y)$$

$$\text{var}(S_I) = \lambda E(Y^2)$$

$$\text{skew}(S_I) = \lambda E(Y^3)$$

Similar formulae can be obtained for S_R by replacing Y with Z .

Note, however that if $F(M) > 0$, as will usually be the case, then Z_i may take the value 0.

Here, $F(x)$ denotes the distribution function of the original claim amount random variable, X .
So:

$$F(M) = P(X \leq M)$$

If this is greater than 0, then there is a non-zero probability that the reinsurer will not be involved in a claim.

In other words, 0 is counted as a possible claim amount for the reinsurer. From a practical point of view, this definition of S_R is rather artificial. The insurer will know the observed value of N , but the reinsurer will probably know only the number of claims above the retention level M , since the insurer may notify the reinsurer only of claims above the retention level.

Example

The annual aggregate claim amount from a risk has a compound Poisson distribution with Poisson parameter 10. Individual claim amounts are uniformly distributed on $(0, 2000)$. The insurer of this risk has effected excess of loss reinsurance with retention level 1,600.

Calculate the mean, variance and coefficient of skewness of both the insurer's and reinsurer's aggregate claims under this reinsurance arrangement.

Solution

Let S_I and S_R be as above. To find $E[S_I]$ calculate $E[Y_i]$. Now:

$$E(Y_i) = \int_0^M x f(x) dx + M P(X_i > M)$$

where $f(x) = 0.0005$ is the $U(0, 2000)$ density function and $M = 1,600$.

This gives:

$$E[Y_i] = \left[\frac{0.0005x^2}{2} \right]_0^M + 0.2M = 960$$

So:

$$E[S_I] = 10E[Y_I] = 9,600$$

To find $\text{var}[S_I]$, we must calculate $E[Y_I^2]$:

$$\begin{aligned} E[Y_I^2] &= \int_0^M x^2 f(x) dx + M^2 P(X_I > M) \\ &= \left[\frac{0.0005x^3}{3} \right]_0^M + 0.2M^2 \\ &= 1,194,666.7 \end{aligned}$$

So:

$$\text{var}[S_I] = 10E[Y_I^2] = 11,946,667$$

To find the coefficient of skewness of the insurer's claims, we must calculate $E[Y_I^3]$:

$$\begin{aligned} E[Y_I^3] &= \int_0^M x^3 f(x) dx + M^3 P(X_I > M) \\ &= \left[\frac{0.0005x^4}{4} \right]_0^M + 0.2M^3 \\ &= 1,638,400,000 \end{aligned}$$

So:

$$E[(S_I - E(S_I))^3] = 10E[Y_I^3] = 16,384,000,000$$

and the coefficient of skewness of S_I is:

$$\frac{16,384,000,000}{11,946,667^{3/2}} = 0.397$$

To find $E[S_R]$, note that the expected annual aggregate claim amount from the risk is $E[S] = \lambda E[X] = 10 \times 1,000 = 10,000$. Then:

$$E[S_R] = 10,000 - E[S_I] = 400$$

To find $\text{var}[S_R]$, calculate $E[Z_i^2]$ from:

$$\begin{aligned} E[Z_i^2] &= \int_M^{2,000} (x - M)^2 f(x) dx \\ &= \int_0^{2,000-M} 0.0005 y^2 dy \quad \text{where } y = x - M \\ &= \left[\frac{0.0005 y^3}{3} \right]_0^{2,000-M} \\ &= 10,666.7 \end{aligned}$$

So:

$$\text{var}[S_R] = 10E[Z_i^2] = 106,667$$

To find the coefficient of skewness of the reinsurer's claims, we need to calculate $E[Z_i^3]$:

$$\begin{aligned} E[Z_i^3] &= \int_M^{2,000} (x - M)^3 f(x) dx \\ &= \int_0^{2,000-M} 0.0005 y^3 dy \quad \text{where } y = x - M \\ &= 3,200,000 \end{aligned}$$

So:

$$E[(S_R - E(S_R))^3] = 10E[Z_i^3] = 32,000,000$$

and the coefficient of skewness of S_R is:

$$\frac{32,000,000}{106,667^{3/2}} = 0.92$$



Question

Calculate the variance of S , the aggregate claim amount before reinsurance for the example above and explain why:

$$\text{var}(S_I) + \text{var}(S_R) \neq \text{var}(S)$$

Solution

We have $\text{var}(S) = 10E(X^2)$, where:

$$E(X^2) = \int_0^{2000} \frac{x^2}{2000} dx = \frac{4,000,000}{3}$$

So $\text{var}(S) = 13,333,333$.

This is not equal to $11,946,667 + 106,667$ because S_I and S_R are not independent.



To simulate the collective risk model with *individual* reinsurance we can combine the R code from Chapters 15, 18 and 19.

For example, to simulate 10,000 values for a reinsurer where claims have a compound Poisson distribution with parameter 1,000 and a gamma claims distribution with $\alpha = 750$ and $\lambda = 0.25$ under *individual* excess of loss with retention 2,500 we would use:

```
set.seed(123)
M <- 2500
n <- rpois(10000, 1000)
sR <- numeric(10000)
for(i in 1:10000)
{ x <- rgamma(n[i], shape=750, rate=0.25)
  z <- pmax(0, x-M)
  sR[i] <- sum(z) }
```

We can now find moments, the coefficient of skewness, probabilities and quantiles as before.

Earlier we mentioned that using $S_R = Z_1 + \dots + Z_N$ is a bit artificial. We now look at an alternative way of modelling the reinsurer's compound claim amount distribution.

The reinsurer's aggregate claims can also be represented by:

$$S_R = W_1 + W_2 + \dots + W_{NR} \quad (20.2)$$

where the random variable NR denotes the actual number of (non-zero) payments made by the reinsurer.

Here:

$$W_i = X_i - M \mid X_i > M$$

For example, suppose that the risk above gave rise to the following eight claim amounts in a particular year:

403 1,490 1,948 443 1,866 1,704 1,221 823

The retention limit is 1,600.

Then in formula (20.1) the observed value of N is 8, and the third, fifth and sixth claims require payments from the reinsurer of 348, 266 and 104 respectively. The reinsurer makes a 'payment' of 0 on the other five claims.

In formula (20.2), the observed value of NR is 3 and the observed values of W_1 , W_2 and W_3 are 348, 266 and 104 respectively. Note that the observed value of S_R is the same (ie 718) under each definition.

W_i has density function:

$$g(w) = \frac{f(x + M)}{1 - F(M)}, \quad w > 0$$

We saw this result in Section 1.2 of [Chapter 18](#). It can also be written as:

$$f_W(w) = \frac{f_X(w + M)}{1 - F_X(M)}$$

To specify the distribution for S_R as given in formula (20.2) the distribution of NR is needed.

In some contexts it may be obvious what this distribution is, but here is a general method for establishing the distribution.

This is found as follows. Define:

$$NR = I_1 + I_2 + \dots + I_N$$

where N denotes the number of claims from the risk (as usual). I_j is an indicator random variable which takes the value 1 if the reinsurer makes a (non-zero) payment on the j th claim, and takes the value 0 otherwise. Thus NR gives the number of payments made by the reinsurer.

From its definition, we see that NR is a compound random variable. However, instead of being the sum of individual claims amounts, NR is a sum of indicator random variables.

Since I_j takes the value 1 only if $X_j > M$:

$$P(I_j = 1) = P(X_j > M) = \pi, \text{ say}$$

and:

$$P(I_j = 0) = 1 - \pi$$

In other words, I_j has a *Binomial*(1, π) distribution.

Further, I_j has MGF:

$$M_{I_j}(t) = \pi e^t + 1 - \pi$$

The formula for the MGF of a binomial distribution is given on page 6 of the *Tables*.

By formula (19.8) in Chapter 19 (the formula for the MGF of a compound random variable), NR has MGF:

$$M_{NR}(t) = M_N(\log M_I(t))$$



Question

If N has a $Poisson(\lambda)$ distribution and $P(X > M) = \frac{1}{2}$, show that NR has a $Poisson(\frac{1}{2}\lambda)$ distribution.

Solution

Here $\pi = \frac{1}{2}$, so:

$$M_I(t) = \frac{1}{2}e^t + \frac{1}{2}$$

and:

$$M_{NR}(t) = M_N[\log M_I(t)] = \exp\{\lambda[M_I(t) - 1]\} = \exp\left\{\lambda\left(\frac{1}{2}e^t - \frac{1}{2}\right)\right\} = \exp\left\{\frac{1}{2}\lambda(e^t - 1)\right\}$$

This is the MGF of the $Poisson(\frac{1}{2}\lambda)$ distribution. By the uniqueness property of MGFs, it follows that $NR \sim Poisson(\frac{1}{2}\lambda)$.

We now continue the above Core Reading example where the annual aggregate claim amount from a risk has a compound Poisson distribution with Poisson parameter 10, individual claim amounts are uniformly distributed on $(0, 2000)$, and the insurer of this risk has effected excess of loss reinsurance with retention level 1,600.

Example

Continuing the above example and using formula (20.2) as the model for S_R , it can be seen that S_R has a compound Poisson distribution with Poisson parameter $0.2 \times 10 = 2$.

Individual claims, W_i , have density function:

$$g(w) = \frac{f(w + M)}{1 - F(M)} = \frac{0.0005}{0.2} = 0.0025, \text{ for } 0 < w < 400$$

ie W_i is uniformly distributed on $(0, 400)$.

Using the formulae for the moments of a continuous uniform distribution from page 13 of the Tables, we have:

$$E[W_i] = 200, E[W_i^2] = 53,333.33 \text{ and } E[W_i^3] = 16,000,000$$

giving the same results as before.

If we multiply these figures by 2 (the Poisson parameter of S_R), we get $E(S_R) = 400$, $\text{var}(S_R) = 106,667$ and $\text{skew}(S_R) = 32,000,000$, which agree with the answers obtained previously.

Thus, there are two ways to specify and evaluate the distribution of S_R .

1.3 Aggregate excess of loss reinsurance

Under an aggregate excess of loss arrangement with retention limit M , the insurer pays all the claims if the total claim amount is less than or equal to the retention limit. The maximum payment made by the insurer is M . So the insurer's aggregate claim payment is:

$$S_I = \begin{cases} S & \text{if } S \leq M \\ M & \text{if } S > M \end{cases}$$

The reinsurer's aggregate claim payment is:

$$S_R = \begin{cases} 0 & \text{if } S \leq M \\ S - M & \text{if } S > M \end{cases}$$

Calculations involving aggregate excess of loss reinsurance are done using a first principles approach.



Question

The annual number of claims from a small group of policies has a Poisson distribution with a mean of 2. Individual claim amounts have the following distribution:

Amount	200	400
Probability	0.7	0.3

Individual claim amounts are independent of each other and are also independent of the number of claims. The insurer has purchased aggregate excess of loss reinsurance with a retention limit of 600.

Calculate the probability that the reinsurer is involved in paying the claims that arise in the next policy year.

Solution

The reinsurer will be involved if the total claim amount is more than 600. Since the total claim amount must be a multiple of 200, the probability is:

$$P(S > 600) = 1 - P(S = 0) - P(S = 200) - P(S = 400) - P(S = 600)$$

The total claim amount will be 0 only if there are no claims. So:

$$P(S=0) = P(N=0) = \frac{e^{-2} 2^0}{0!} = e^{-2}$$

Using the assumption that individual claim amounts are independent of the number of claims, we have:

$$P(S=200) = P(N=1, X_1=200) = P(N=1)P(X_1=200) = \frac{e^{-2} 2^1}{1!} \times 0.7 = 1.4 e^{-2}$$

$$\begin{aligned} P(S=400) &= P(N=2, X_1=200, X_2=200) + P(N=1, X_1=400) \\ &= \frac{e^{-2} 2^2}{2!} \times 0.7^2 + \frac{e^{-2} 2^1}{1!} \times 0.3 \\ &= 1.58 e^{-2} \end{aligned}$$

and:

$$\begin{aligned} P(S=600) &= P(N=3, X_1=200, X_2=200, X_3=200) \\ &\quad + P(N=2, X_1=200, X_2=400) + P(N=2, X_1=400, X_2=200) \\ &= \frac{e^{-2} 2^3}{3!} \times 0.7^3 + 2 \times \frac{e^{-2} 2^2}{2!} \times 0.7 \times 0.3 \\ &= 1.29733 e^{-2} \end{aligned}$$

So:

$$P(S > 600) = 1 - e^{-2} (1 + 1.4 + 1.58 + 1.29733) = 0.28579$$



We can also simulate the collective risk model with *aggregate* reinsurance. For example to simulate 10,000 values for a reinsurer where claims have a compound Poisson distribution with parameter 1,000 and a gamma claims distribution with $\alpha = 750$ and $\lambda = 0.25$ under *aggregate* excess of loss with retention 3,000,000 we would take our S from Section 3.4 of [Chapter 19](#) and then use:

```
sR <- pmax(0, s-3000000)
```

2 The individual risk model

Under this model a portfolio consisting of a fixed number of risks is considered. It will be assumed that:

- these risks are independent
- claim amounts from these risks are not (necessarily) identically distributed random variables
- the number of risks does not change over the period of insurance cover.

As before, aggregate claims from this portfolio are denoted by S . So:

$$S = Y_1 + Y_2 + \dots + Y_n$$

where Y_j denotes the claim amount under the j th risk and n denotes the number of risks. It is possible that some risks will not give rise to claims. Thus, some of the observed values of $\{Y_j\}_{j=1}^n$ may be 0.

In fact in most forms of insurance most policies would not give rise to any claims during a given year.

This approach is referred to as an *individual risk model* because it is considering the claims arising from each individual policy.

For each risk, the following assumptions are made:

- the number of claims from the j th risk, N_j , is either 0 or 1 (20.3)

- the probability of a claim from the j th risk is q_j . (20.4)

If a claim occurs under the j th risk, the claim amount is denoted by the random variable X_j . Let $F_j(x)$, μ_j and σ_j^2 denote the distribution function, mean and variance of X_j respectively.

Assumption (20.3) is very restrictive. It means that a maximum of one claim from each risk is allowed for in the model. This includes risks such as one-year term assurance (since a policyholder can only die once), but excludes many types of general insurance policy. For example, there is no restriction on the number of claims that could be made in a policy year under household contents insurance.

There are three important differences between this model and the collective risk model:

- (1) The number of risks in the portfolio has been specified. In the collective risk model, there was no need to specify this number, nor to assume that it remained fixed over the period of insurance cover (not even when it was assumed that $N \sim \text{Bin}(n, q)$).

On the other hand we could argue that there was an implicit assumption of a constant number of risks in the very fact that we were using a binomial distribution to model the number of claims.

- (2) The number of claims from each individual risk has been restricted. There was no such restriction in the collective risk model.

- (3) It is assumed that individual risks are independent. In the collective risk model it was individual claim amounts that were independent.

The contrast here is between the occurrence of claims and the size of claims.

Assumptions (20.3) and (20.4) say that $N_j \sim \text{Bin}(1, q_j)$. Thus, the distribution of Y_j is compound binomial, with individual claim amount random variable X_j . From formulae (19.15) and (19.16) in Chapter 19 (for the mean and variance of a compound random variable) it follows immediately that:

$$E[Y_j] = q_j \mu_j \quad (20.5)$$

$$\text{var}[Y_j] = q_j \sigma_j^2 + q_j(1 - q_j) \mu_j^2 \quad (20.6)$$

S is the sum of n independent compound binomial random variables. The distribution of S can be stated only when the compound binomial variables are identically distributed, as well as independent. It is possible, but complicated, to compute the distribution function of S under certain conditions.

However, it is easy to find the mean and variance of S :

$$E[S] = E\left[\sum_{j=1}^n Y_j\right] = \sum_{j=1}^n E[Y_j] = \sum_{j=1}^n q_j \mu_j \quad (20.7)$$

The assumption that individual risks are independent is needed to write:

$$\text{var}[S] = \text{var}\left[\sum_{j=1}^n Y_j\right] = \sum_{j=1}^n \text{var}[Y_j] = \sum_{j=1}^n (q_j \sigma_j^2 + q_j(1 - q_j) \mu_j^2) \quad (20.8)$$

In the special case when $\{Y_j\}_{j=1}^n$ is a sequence of identically distributed, as well as independent, random variables, then for each policy the values of q_j , μ_j and σ_j^2 are identical, say q , μ and σ^2 . Since $F_j(x)$ is independent of j , we can refer to it simply as $F(x)$. Hence, S is compound binomial, with binomial parameters n and q , and individual claims have distribution function $F(x)$. In this special case, it reduces to the collective risk model, and it can be seen from (20.7) and (20.8) that:

$$E[S] = nq\mu$$

$$\text{var}[S] = nq\sigma^2 + nq(1 - q)\mu^2$$

which correspond to (19.15) and (19.16) respectively in Chapter 19.



Question

The probability of a claim arising on any given policy in a portfolio of 1,000 one-year term assurance policies is 0.004. Claim amounts have a $\text{Gamma}(5, 0.002)$ distribution. Calculate the mean and variance of the aggregate claim amount.

Solution

We have:

$$\mu_j = \frac{5}{0.002} = 2,500 \quad \text{and:} \quad \sigma_j^2 = \frac{5}{0.002^2} = 1,250,000$$

So the mean and variance of the aggregate claim amount are:

$$E(S) = nq\mu = 1,000 \times 0.004 \times 2,500 = 10,000$$

and:

$$\begin{aligned} \text{var}(S) &= nq\sigma^2 + nq(1-q)\mu^2 \\ &= 1,000 \times \left[0.004 \times 1,250,000 + 0.004 \times 0.996 \times 2,500^2 \right] \\ &= (\text{£}5,468)^2 \end{aligned}$$

We can use R to simulate the total claim amount payable under the individual risk model.



Suppose we have n life policies, with the probabilities of death for each policy contained in the vector `q` and simulated claim amounts for each policy contained in the vector `claim`. Then:

```
S <- q*claim
```

We can now find moments, the coefficient of skewness, probabilities and quantiles as before, and also apply reinsurance if appropriate.

3 Parameter variability / uncertainty

This section forms part of the Core Reading, but does not address any specific syllabus objectives. However, the material here provides useful practice in applying the models we have studied.

3.1 Introduction

So far risk models have been studied assuming that the parameters, that is the moments and in some cases even the distributions, of the number of claims and of the amount of individual claims are known with certainty. In general, these parameters would not be known but would have to be estimated from appropriate sets of data. In this section it will be seen how the models introduced earlier can be extended to allow for parameter uncertainty / variability. This will be done by looking at a series of examples. Most, but not all, of these examples will consider uncertainty in the claim number distribution since this, rather than the individual claim amount distribution, has received more attention in the actuarial literature. All the examples will be based on claim numbers having a Poisson distribution.

3.2 Variability in a heterogeneous portfolio

Consider a portfolio consisting of n independent policies. The aggregate claims from the i th policy are denoted by the random variable S_i , where S_i has a compound Poisson distribution with parameters λ_i , and the CDF of the individual claim amounts distribution is $F(x)$. Notice that, for simplicity, the CDF of the distribution of individual claim amounts, $F(x)$, is assumed to be identical for all the policies.

In this example the CDF of individual claim amounts, ie $F(x)$, is assumed to be known but the values of the Poisson parameters, ie the λ_i s, are not known. In this subsection the λ_i s are assumed to be (sample values of) independent random variables with the same (known) distribution. In other words $\{\lambda_i\}_{i=1}^n$ is treated as a set of independent and identically distributed random variables with a known distribution. This means that if a policy is chosen at random from the portfolio it is assumed that the Poisson parameter for the policy is not known but that probability statements can be made about it. For example, 'there is a 50% chance that its Poisson parameter lies between 3 and 5'. It is important to understand that the Poisson parameter for a policy chosen from the portfolio is a fixed number; the problem is that this number is not known.

Example

Suppose that the Poisson parameters of policies in a portfolio are not known but are equally likely to be 0.1 or 0.3.

- (i) Find the mean and variance (in terms of m_1 and m_2) of the aggregate claims from a policy chosen at random from the portfolio.
- (ii) Find the mean and variance (in terms of m_1 , m_2 and n) of the aggregate claims from the whole portfolio.

It may be helpful to think of this as a model of part of a motor insurance portfolio. The policies in the whole portfolio have been subdivided according to their values for rating factors such as 'age of driver', 'type of car' and even 'past claims experience'. The policies in the part of the portfolio being considered have identical values for these rating factors. However, there are some factors, such as 'driving ability', that cannot easily be measured and so they cannot be taken explicitly into account. It is supposed that some of the policyholders in this part of the portfolio are 'good' drivers and the remainder are 'bad' drivers. The individual claim amount distribution is the same for all drivers but 'good' drivers make fewer claims (0.1 *pa* on average) than 'bad' drivers (0.3 *pa* on average). It is assumed that it is known, possibly from national data, that a policyholder in this part of the portfolio is equally likely to be a 'good' driver or a 'bad' driver but that it cannot be known whether a particular policyholder is a 'good' driver or a 'bad' driver.

Solution

Let λ_i , $i = 1, 2, \dots, n$ be the Poisson parameter of the i th policy in the portfolio. $\{\lambda_i\}_{i=1}^{\infty}$ is regarded as a set of independent and identically distributed random variables, each with the following distribution:

$$P(\lambda_i = 0.1) = 0.5$$

$$P(\lambda_i = 0.3) = 0.5$$

From this:

$$E[\lambda_i] = 0.2$$

$$\text{var}[\lambda_i] = 0.01$$

- (i) The moments of S_i can be calculated by conditioning on the value of λ_i . Since $S_i | \lambda_i$ has a straightforward compound Poisson distribution, formulae (19.8) and (19.9) in [Chapter 19](#) can be used to write:

$$E[S_i] = E[E[S_i | \lambda_i]] = E[\lambda_i m_1] = 0.2 m_1$$

$$\begin{aligned} \text{var}[S_i] &= E[\text{var}[S_i | \lambda_i]] + \text{var}[E[S_i | \lambda_i]] \\ &= E[\lambda_i m_2] + \text{var}[\lambda_i m_1] \\ &= 0.2 m_2 + 0.01 m_1^2 \end{aligned}$$

- (ii) The random variables $\{S_i\}_{i=1}^n$ are independent and identically distributed, each with the distribution of S_i given in part (i). Hence, the result in (i) above can be used to write:

$$E\left[\sum_{i=1}^n S_i\right] = n E[S_i] = 0.2 n m_1$$

$$\text{var}\left[\sum_{i=1}^n S_i\right] = n \text{var}[S_i] = 0.2 n m_2 + 0.01 n m_1^2$$

Example

Suppose the Poisson parameters for individual policies are drawn from a gamma distribution with parameters α and δ . Find the distribution of the number of claims from a policy chosen at random from the portfolio.

Solution

Let N_i denote the number of claims from the i th policy in the portfolio and let λ_i be its Poisson parameter. Then N_i has a Poisson distribution with parameter λ_i but the problem is that (by assumption) the value of λ_i is not known. What is known is the distribution from which λ_i has been chosen.

The problem can be summarised as follows:

Given that:

$$N_i | \lambda_i \sim \text{Poisson}(\lambda_i) \text{ and } \lambda_i \sim \text{Gamma}(\alpha, \delta)$$

find the marginal distribution of N_i .

The marginal distribution of N_i is its unconditional distribution. In this example, N_i can only take whole number values, so it is a discrete random variable. To determine its marginal distribution, we need to derive a formula for the unconditional probability $P(N_i = x)$.

This problem can be solved by removing the conditioning in the usual way.

Recall that if X and Y are discrete random variables, then the unconditional probability $P(X = x)$ is given by:

$$P(X = x) = \sum_y P(X = x, Y = y) = \sum_y P(X = x | Y = y)P(Y = y)$$

In this example X is being replaced by N_i and Y is being replaced by λ_i . Since λ_i is a continuous random variable, we turn the sum into an integral and formula becomes:

$$P(N_i = x) = \int_{\lambda_i} P(N_i = x | \lambda_i) f(\lambda_i) d\lambda_i$$

For $x = 0, 1, 2, \dots$:

$$\begin{aligned} P(N_i = x) &= \int_0^\infty \exp\{-\lambda\} \frac{\lambda^x}{x!} \frac{\delta^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} \exp\{-\delta\lambda\} d\lambda \\ &= \frac{\delta^\alpha}{\Gamma(\alpha)x!} \int_0^\infty \exp\{-\lambda(\delta+1)\} \lambda^{x+\alpha-1} d\lambda \end{aligned}$$

Evaluate the integral by comparing the integrand with a gamma density.

We can make the integrand look like the PDF of the $\text{Gamma}(x + \alpha, \delta + 1)$ distribution by inserting a factor of $\frac{(\delta + 1)^{x+\alpha}}{\Gamma(x + \alpha)}$ inside the integral. We need to compensate for doing this by inserting a factor of $\frac{\Gamma(x + \alpha)}{(\delta + 1)^{x+\alpha}}$ outside the integral. This gives:

$$\begin{aligned} P(N_i = x) &= \frac{\delta^\alpha}{\Gamma(\alpha)x!} \frac{\Gamma(x + \alpha)}{(\delta + 1)^{x+\alpha}} \int_0^\infty \frac{(\delta + 1)^{x+\alpha}}{\Gamma(x + \alpha)} \lambda^{x+\alpha-1} e^{-\lambda(\delta+1)} d\lambda \\ &= \frac{\delta^\alpha}{\Gamma(\alpha)x!} \frac{\Gamma(x + \alpha)}{(\delta + 1)^{x+\alpha}} \int_0^\infty f(\lambda) d\lambda \quad \text{where } \lambda \sim \text{Gamma}(x + \alpha, \delta + 1) \end{aligned}$$

The integral in the line above is 1 (as we are integrating a PDF over all possible values of the random variable).

So:

$$P(N_i = x) = \frac{\delta^\alpha}{\Gamma(\alpha)x!} \frac{\Gamma(x + \alpha)}{(\delta + 1)^{x+\alpha}}$$

which shows that the marginal distribution of N_i is negative binomial with parameters α and $\frac{\delta}{\delta + 1}$.

3.3 Variability in a homogeneous portfolio

Now a different example is considered. Suppose, as before, there is a portfolio of n policies. The aggregate claims from a single policy have a compound Poisson distribution with parameters λ , and the CDF of the individual claim amounts random variable is $F(x)$. The Poisson parameters are the same for all policies in the portfolio. If the value of λ were known, the aggregate claims from different policies would be independent of each other. It is assumed that the value of λ is not known, possibly because it changes from year to year, but that there is some indication of the probability that λ will be in any given range of values. As in the previous example, it is assumed for simplicity that there is no uncertainty about the moments or distribution of the individual claim amounts, ie about $F(x)$. The uncertainty about the value of λ can be modelled by regarding λ as a random variable (with a known distribution).

Example

Suppose that the Poisson parameter, λ , will be equal to 0.1 or to 0.3 with equal probability.

- (i) Calculate the mean and variance (in terms of m_1 and m_2) of the aggregate claims from a policy chosen at random from the portfolio.
- (ii) Calculate the mean and variance (in terms of m_1 , m_2 and n) of the aggregate claims from the whole portfolio.

Solution

Using the same notation as before let S_i denote the aggregate claims from the i th policy in the portfolio. The situation can be summarised as follows:

The random variables $\{S_i | \lambda\}_{i=1}^n$ are independent and identically distributed, each with a compound Poisson distribution with parameters λ and $F(x)$. The random variable λ has the following distribution:

$$P(\lambda = 0.1) = 0.5$$

$$P(\lambda = 0.3) = 0.5$$

(i) Conditioning on the value of λ :

$$E[S_i] = E[E(S_i | \lambda)] = E[\lambda m_1] = 0.2 m_1$$

$$\begin{aligned} \text{var}[S_i] &= E[\text{var}(S_i | \lambda)] + \text{var}[E(S_i | \lambda)] = E[\lambda m_2] + \text{var}[\lambda m_1] \\ &= 0.2 m_2 + 0.01 m_1^2 \end{aligned}$$

$$(ii) \quad E\left[\sum_{i=1}^n S_i\right] = n E[S_1] = 0.2 n m_1$$

(since $\{S_i\}_{i=1}^n$ are identically distributed)

$$\begin{aligned} \text{var}\left[\sum_{i=1}^n S_i\right] &= E\left[\text{var}\left(\sum_{i=1}^n S_i | \lambda\right)\right] + \text{var}\left[E\left(\sum_{i=1}^n S_i | \lambda\right)\right] \\ &= E[n \lambda m_2] + \text{var}[n \lambda m_1] \\ &= 0.2 n m_2 + 0.01 n^2 m_1^2 \end{aligned}$$

Note that $S_1 | \lambda, \dots, S_n | \lambda$ are independent but $S_1, \dots, S_n$ are not unconditionally independent

(since they all depend on the value of λ), so $\text{var}\left(\sum_{i=1}^n S_i\right) \neq \sum_{i=1}^n \text{var}(S_i)$.

It is useful to compare the answers to the above example with those to the first example in the Section 3.2. The values of the mean are in all cases the same, as are the variances when a single policy is considered (part (i)). The difference occurs when variances for more than one policy are considered (part (ii)), in which case the second example gives the greater variance. It is important to understand the differences (and the similarities) between the two examples. A practical situation where the second example could be appropriate would be a portfolio of policies insuring buildings in a certain area. The number of claims could depend on, among other factors, the weather during the year; an unusually high number of storms resulting in a high expected number of claims (ie a high value of λ) and vice versa for all the policies together.

3.4 Variability in claim numbers and claim amounts and parameter uncertainty

This section contains two more examples. The first is a rather complicated example involving uncertainty over claim amounts as well as claim numbers.

Example

An insurance company models windstorm claims under household insurance policies using the following assumptions.

The number of storms arising each year, K , is assumed to have a Poisson distribution with parameter λ .

The number of claims arising from the i th storm, N_i , $i = 1, 2, \dots, K$, is assumed to have a Poisson distribution with parameter Θ_i .

The parameters Θ_i , $i = 1, 2, \dots, K$, are assumed to be independent and identically distributed random variables, with $E(\Theta_i) = n$ and $\text{var}(\Theta_i) = s_1^2$.

The amount of the j th claim arising from the i th storm, X_{ij} , $j = 1, 2, \dots, N_i$, has a lognormal distribution with parameters μ_i and σ^2 , where σ^2 is assumed to be known. The mean claim amounts, $\Lambda_i = \exp(\mu_i + \sigma^2 / 2)$ are assumed to be independent and identically distributed random variables with mean p and variance s_2^2 .

It is also assumed that Θ_i and Λ_i are independent.

- (i) Show that $E[X_{ij}] = p$ and $\text{var}[X_{ij}] = \exp\{\sigma^2\} (p^2 + s_2^2) - p^2$.
- (ii) Let S_i denote aggregate claims outgo from the i -th storm, so that $S_i | \{\Theta_i, \Lambda_i\}$ is a compound Poisson random variable. Show that:

$$E[S_i] = np$$

and:

$$\text{var}[S_i] = (p^2 + s_2^2) (n^2 + s_1^2 + n \exp\{\sigma^2\}) - n^2 p^2$$

- (iii) Find expressions for the mean and variance of the annual aggregate claims outgo from all storms.

Solution

$$(i) \quad E[X_{ij}] = E[E(X_{ij} | \Lambda_i)] = E[\Lambda_i] = p$$

$$\begin{aligned} \text{var}[X_{ij}] &= E[\text{var}(X_{ij} | \Lambda_i)] + \text{var}[E(X_{ij} | \Lambda_i)] \\ &= E[\Lambda_i^2 (\exp\{\sigma^2\} - 1)] + \text{var}(\Lambda_i) \\ &= (p^2 + s_2^2) (\exp\{\sigma^2\} - 1) + s_2^2 \\ &= (p^2 + s_2^2) \exp\{\sigma^2\} - p^2 \end{aligned}$$

- (ii) $E[S_i] = E[E(S_i | \Theta_i, \Lambda_i)] = E[\Theta_i \Lambda_i] = np$ since Θ_i and Λ_i are independent.

Now, since $S_i | \{\Theta_i, \Lambda_i\}$ has a compound Poisson distribution:

$$\text{var}[S_i | \Theta_i, \Lambda_i] = \Theta_i E[X_{ij}^2 | \Lambda_i] = \Theta_i (\Lambda_i^2 \exp\{\sigma^2\})$$

and so:

$$E[\text{var}(S_i | \Theta_i, \Lambda_i)] = n(p^2 + s_2^2) \exp\{\sigma^2\}$$

Also:

$$\begin{aligned} \text{var}[E(S_i | \Theta_i, \Lambda_i)] &= \text{var}[\Theta_i \Lambda_i] = E[\Theta_i^2 \Lambda_i^2] - n^2 p^2 \\ &= (n^2 + s_1^2)(p^2 + s_2^2) - n^2 p^2 \end{aligned}$$

Putting these last two results together:

$$\text{var}[S_i] = (n^2 + s_1^2)(p^2 + s_2^2) - n^2 p^2 + n(p^2 + s_2^2) \exp\{\sigma^2\}$$

- (iii) Let R be a random variable denoting the annual aggregate claims outgo from all storms. Then R can be written:

$$R = \sum_{i=1}^K S_i$$

where K has a Poisson distribution and the random variables S_i are IID (independent and identically distributed).

Hence, R has a compound Poisson distribution and so:

$$E[R] = \lambda E(S_i) = \lambda n p$$

$$\begin{aligned} \text{var}[R] &= \lambda E(S_i^2) = \lambda (\text{var}[S_i] + E[S_i]^2) \\ &= \lambda (p^2 + s_2^2)(n^2 + s_1^2 + n \exp\{\sigma^2\}) \end{aligned}$$

Example

Each year an insurance company issues a number of household contents insurance policies, for each of which the annual premium is £80. The aggregate annual claims from a single policy have a compound Poisson distribution; the Poisson parameter is 0.4 and individual claim amounts have a gamma distribution with parameters α and λ . The expense involved in settling a claim is a random variable uniformly distributed between £50 and £ b (>£50). The amount of the expense is independent of the amount of the associated claim. The random variable S represents the total aggregate claims and expenses in one year from this portfolio. It may be assumed that S has approximately a normal distribution.

(i) Suppose that:

$$\alpha = 1; \lambda = 0.01; b = 100$$

Show that the company must sell at least 884 policies in a year to be at least 99% sure that the premium income will exceed the claims and expenses outgo.

(ii) Now suppose that the values of α , λ and b are not known with certainty but could be anywhere in the following ranges:

$$0.95 \leq \alpha \leq 1.05; 0.009 \leq \lambda \leq 0.011; 90 \leq b \leq 110$$

By considering what, for the insurance company, would be the worst possible combination of values for α , λ and b , calculate the number of policies the company must sell to be at least 99% sure that the premium income will exceed the claims and expenses outgo.

Solution

Let X_i be the amount of the i th claim and Y_i be the amount of the associated expense. Let N be the total number of claims from the portfolio and let n be the number of policies in the portfolio. Then N has a Poisson distribution with parameter $0.4n$ and S can be written:

$$S = \sum_{i=1}^N (X_i + Y_i)$$

where $\{X_i + Y_i\}_{i=1}^{\infty}$ is a sequence of independent and identically distributed random variables, independent of N . From this it can be seen that S has a compound Poisson distribution with $X_i + Y_i$ representing the 'amount of the i th individual claim'. Standard results can now be used to write down the following formulae for the moments of S :

$$E[S] = 0.4n E[X_i + Y_i]$$

$$\text{var}[S] = 0.4n E[(X_i + Y_i)^2] = 0.4n (E[X_i^2] + 2E[X_i Y_i] + E[Y_i^2])$$

In terms of α , λ and b , the moments of X_i and Y_i are as follows:

$$E[X_i] = \alpha / \lambda$$

$$E[Y_i] = (b + 50) / 2$$

$$E[X_i^2] = \alpha(\alpha + 1) / \lambda^2$$

$$E[Y_i^2] = (b^2 + 50b + 2,500) / 3$$

$$E[X_i Y_i] = E[X_i] E[Y_i]$$

where the final relationship follows from the independence of X_i and Y_i .

(i) Now put:

$$\alpha = 1; \lambda = 0.01; b = 100$$

into these formulae to show that:

$$E[S] = 70n \text{ and } \text{var}[S] = 127.80^2 n$$

Hence, S has approximately a normal distribution with mean $70n$ and standard deviation $127.80\sqrt{n}$. The premium income is $80n$ and the smallest value of n is required such that:

$$P(S < 80n) \geq 0.99$$

Standardising S in the usual way for a normal distribution, this becomes:

$$P\left[\frac{S - 70n}{127.80\sqrt{n}} < \frac{80n - 70n}{127.80\sqrt{n}}\right] \geq 0.99$$

The upper 99% point of a standard normal distribution is 2.326 and so the condition for n is:

$$\frac{80n - 70n}{127.80\sqrt{n}} \geq 2.326$$

which gives:

$$n \geq 883.7$$

(or $n \geq 884$ to the next higher integer).

(ii) For the insurance company, the worst possible combination of values for α , λ and b is the combination which gives the highest possible values for $E[S]$ and $\text{var}[S]$. To see this, let μ and σ denote the mean and the standard deviation of aggregate claims and expenses from a single policy. Both μ and σ will be functions of α , λ and b and:

$$E[S] = n\mu \text{ and } \text{var}[S] = n\sigma^2$$

Following the same steps as in part (i), the condition for n is:

$$\frac{(80 - \mu)\sqrt{n}}{\sigma} \geq 2.326$$

which becomes:

$$n \geq [2.326\sigma / (80 - \mu)]^2$$

Hence, the highest value of n results from the highest values for μ and σ (provided the highest value for μ is less than 80). Now note that:

$$\mu = 0.4E[X_i + Y_i] \quad \text{and} \quad \sigma^2 = 0.4E[(X_i + Y_i)^2]$$

From the formulae for the moments of X_i and Y_i given above, μ and σ are maximised when α and b are as large as possible and λ is as small as possible, ie when:

$$\alpha = 1.05; \quad \lambda = 0.009; \quad b = 110$$

This combination of values gives:

$$\mu = 78.67 \quad \text{and} \quad \sigma = 144.14$$

so that n must be at least 63,546 for the insurance company to be at least 99% sure that premium income will exceed claims and expenses outgo.

Chapter 20 Summary

Collective risk model with reinsurance

In the collective risk model, individual claims can be subject to a reinsurance agreement, either proportional or excess of loss.

Under the collective risk model, the aggregate claim amount S is given by:

$$S = X_1 + X_2 + \cdots + X_N$$

where X_i is the amount of the i th claim and N is the total number of claims.

If reinsurance is in place, the insurer's aggregate claims net of reinsurance can be represented as:

$$S_I = Y_1 + Y_2 + \cdots + Y_N$$

where Y_i is the amount of the i th claim paid by the insurer and N is defined as above. S_I is a compound random variable and:

$$E(S_I) = E(N)E(Y)$$

$$\text{var}(S_I) = E(N)\text{var}(Y) + \text{var}(N)[E(Y)]^2$$

$$M_{S_I}(t) = M_N[\ln M_Y(t)]$$

The reinsurer's aggregate claims can be represented as:

$$S_R = Z_1 + Z_2 + \cdots + Z_N$$

where Z_i is the amount of the i th claim paid by the reinsurer and N is defined as above. S_R is a compound random variable and:

$$E(S_R) = E(N)E(Z)$$

$$\text{var}(S_R) = E(N)\text{var}(Z) + \text{var}(N)[E(Z)]^2$$

$$M_{S_R}(t) = M_N[\ln M_Z(t)]$$

Under individual excess of loss reinsurance, some of the claims may fall below the retention level M . If this is the case, then some of the Z_i will be zero. An alternative way of expressing the reinsurer's aggregate claims is as:

$$S_R = W_1 + W_2 + \cdots + W_{NR}$$

where $W_i = Z_i \mid Z_i > 0$ and NR is the number of non-zero claims, ie the number of claims in which the reinsurer is involved.

Under an aggregate excess of loss arrangement with retention limit M , the maximum payment made by the insurer is M . The insurer's aggregate claim payment is:

$$S_I = \begin{cases} S & \text{if } S \leq M \\ M & \text{if } S > M \end{cases}$$

The reinsurer's aggregate claim payment is:

$$S_R = \begin{cases} 0 & \text{if } S \leq M \\ S - M & \text{if } S > M \end{cases}$$

Individual risk model

The individual risk model considers the payments made under each risk (*eg* policy) separately. The model assumes that:

- the number of risks is fixed
- the risks are independent
- claim amounts from these risks are not necessarily IID
- N_j , the number of claims from the j th risk is either 0 or 1.

For a portfolio containing n risks, the aggregate claim amount is given by:

$$S = Y_1 + Y_2 + \dots + Y_n$$

where Y_j denotes the aggregate claims from risk j . Since each Y_j is the sum of a random number (0 or 1) of random claim amounts, each Y_j has a compound binomial distribution.

Suppose that q_j is the probability of a claim from the j th risk. If a claim arises from the j th risk, suppose that the claim amount random variable is X_j . Then:

$$E(S) = \sum_{j=1}^n q_j \mu_j$$

$$\text{var}(S) = \sum_{j=1}^n \left[q_j \sigma_j^2 + q_j(1-q_j) \mu_j^2 \right]$$

$$M_S(t) = \prod_{j=1}^n \left[q_j M_{X_j}(t) + (1-q_j) \right]$$

where $\mu_j = E(X_j)$ and $\sigma_j^2 = \text{var}(X_j)$.

If, for a group of n risks, the probability of a claim is fixed and the claim amounts are IID random variables, then the individual risk model is equivalent to a collective risk model where S has a compound binomial distribution with $N \sim \text{Bin}(n, q)$.



Chapter 20 Practice Questions

- 20.1 The annual aggregate claims from a risk have a compound Poisson distribution with parameter 250. Individual claim amounts have a Pareto distribution with parameters $\alpha = 4$ and $\lambda = 900$. The insurer effects proportional reinsurance with a retained proportion of 75%.

Calculate the variances of the total amounts paid by the insurer and by the reinsurer.

- 20.2 The aggregate claims from a risk have a compound Poisson distribution with parameter μ . Individual claim amounts (in £) have a Pareto distribution with parameters $\alpha = 3$ and $\lambda = 1,000$.

The insurer of this risk calculates the premium using a premium loading factor of 0.2 (ie it charges 20% in excess of the risk premium).

The insurer is considering effecting individual excess of loss reinsurance with retention limit £1,000. The reinsurance premium would be calculated using a premium loading factor of 0.3.

The insurer's profit is defined to be the premium charged by the insurer less the reinsurance premium and less the claims paid by the insurer, net of reinsurance.

- (i) Show that the insurer's expected profit before reinsurance is 100μ .
- (ii) Calculate the insurer's expected profit after effecting the reinsurance, and hence find the percentage reduction in the insurer's expected profit.
- (iii) Calculate the percentage reduction in the standard deviation of the insurer's profit as a result of effecting the reinsurance.

- 20.3 Aggregate annual claims from a portfolio of general insurance policies have a compound Poisson distribution with Poisson parameter 20. Individual claim amounts have a uniform distribution over the interval $(0, 200)$. Excess of loss reinsurance is arranged so that the expected amount paid by the insurer on any claim is 50.

Calculate the variance of the aggregate annual claims paid by the insurer.

- 20.4 A portfolio of policies consists of one-year term assurances on 100 lives aged exactly 30 and 200 lives aged exactly 40. The probability of a claim during the year on any one of the lives is 0.0004 for the 30 year olds and 0.001 for the 40 year olds.

If the sum assured on a life aged x is uniformly distributed between $1,000(x - 10)$ and $1,000(x + 10)$, calculate the variance of the aggregate claims from this portfolio during the year (assuming that policies are independent with regard to claims).

- 20.5 The number of claims from a given portfolio has a Poisson distribution with a mean of 1.5 per month. Individual claim amounts have the following distribution:

Amount	200	300
Probability	0.65	0.35

An aggregate reinsurance contract has been arranged so that the insurer pays no more than 400 per month in total.

Assuming that the individual claim amounts are independent of each other and are also independent of the number of claims, calculate the expected aggregate monthly claim amounts for the insurer and the reinsurer.

- 20.6 A portfolio consists of 500 independent risks. For the i th risk, with probability $1 - q_i$ there are no claims in one year, and with probability q_i there is exactly one claim ($0 < q_i < 1$). For all risks, if there is a claim, it has mean μ , variance σ^2 and moment generating function $M(t)$. Let T be the total amount claimed on the whole portfolio in one year.

Exam style

- (i) Determine the mean and variance of T . [4]

The amount claimed in one year on risk i is approximated by a compound Poisson random variable with Poisson parameter q_i and claims with the same mean μ , the same variance σ^2 , and the same moment generating function $M(t)$ as above. Let $\tilde{T}$ denote the total amount claimed on the whole portfolio in one year in this approximate model.

- (ii) Determine the mean and variance of $\tilde{T}$, and compare your answers to those in part (i). [4]

Assume that $q_i = 0.02$ for all i , and if a claim occurs, it is of size μ with probability one.

- (iii) Derive the moment generating function of T , and show that T has a compound binomial distribution. [2]

- (iv) Determine the moment generating function of the approximating $\tilde{T}$, and show that $\tilde{T}$ has a compound Poisson distribution. [2]

[Total 12]

20.7

Exam style

A company is analysing the number of accidents that occur each year on the factory floor. It believes that the number of accidents per year N has a geometric distribution with parameter 0.8, so that:

$$P(N = n) = 0.8 \times 0.2^n, \quad n = 0, 1, 2, \dots$$

For each accident, the number of employees injured is Y , where $Y = X + 1$, and X is believed to have a Poisson distribution with parameter 2.2.

The company has taken out an insurance policy, which provides a benefit of £1,000 to each injured employee, up to a maximum of three employees per accident, irrespective of the level of injury. There is no limit on the number of accidents that may be claimed for in a year.

- (i) Show that $E(S) = 0.634$ and $\text{var}(S) = 2.125$, where S is the total number of employees claiming benefit in a year under this policy. [7]
- (ii) Hence find the mean and variance of the aggregate amount paid out under this policy in a year. [1]

[Total 8]

20.8

Exam style

An insurance company offers accident insurance for employees. A total of 650 policies have been issued split between two categories of employees. The first category contains 400 policies, and claims occur on each policy according to a Poisson process at a rate of one claim per 20 years, on average. In this category all claim amounts are £3,000. In the second category, claims occur on each policy according to a Poisson process at a rate of one claim per 10 years, on average. In this category, the claim amount is either £2,000 or £3,000 with probabilities 0.4 and 0.6, respectively. All policies are assumed to be independent. Let S denote the aggregate annual claims from the portfolio.

- (i) Calculate the mean, variance and coefficient of skewness of S . [4]
- (ii) Using the normal distribution as an approximation to the distribution of S , calculate Y such that the probability of S exceeding Y is 10%. [3]
- (iii) The insurance company decides to effect reinsurance cover with aggregate retention £100,000, so that the insurance company then pays out no more than this amount in claims each year. In the year following the inception of this reinsurance, the numbers of policies in each of the two groups remains the same but, because of changes in the employment conditions of which the company was unaware, the probability of a claim in group 2 falls to zero. Using the normal distribution as an approximation to the distribution of S , calculate the probability of a claim being made on the reinsurance treaty. [3]

[Total 10]

The solutions start on the next page so that you can
separate the questions and solutions.



Chapter 20 Solutions

20.1 The mean and variance of the gross claim amounts are:

$$E(X) = \frac{\lambda}{\alpha - 1} = \frac{900}{3} = 300$$

$$\text{var}(X) = \frac{\alpha \lambda^2}{(\alpha - 1)^2 (\alpha - 2)} = \frac{4 \times 900^2}{3^2 \times 2} = 180,000$$

So the mean and variance of the net claims for the direct insurer and the reinsurer are:

$$E(Y) = 0.75 \times 300 = 225$$

$$\text{var}(Y) = 0.75^2 \times 180,000 = 101,250$$

$$E(Z) = 0.25 \times 300 = 75$$

$$\text{var}(Z) = 0.25^2 \times 180,000 = 11,250$$

Using the formula for the variance of a compound Poisson random variable, the variances of the aggregate claim payments made by the insurer and the reinsurer are:

$$\text{var}(S_I) = \lambda E[Y^2] = \lambda [\text{var}(Y) + [E(Y)]^2] = 250[101,250 + 225^2] = 37,968,750$$

$$\text{var}(S_R) = \lambda E[Z^2] = \lambda [\text{var}(Z) + [E(Z)]^2] = 250[11,250 + 75^2] = 4,218,750$$

20.2 (i) **Expected profit before reinsurance**

We have:

$$E(X) = \frac{\lambda}{\alpha - 1} = 500$$

So the expected aggregate claim amount is:

$$E(S) = 500\mu$$

The insurer's premium income is:

$$1.2E(S) = 1.2 \times 500\mu = 600\mu$$

So the expected profit before reinsurance is:

$$600\mu - 500\mu = 100\mu$$

(ii) **Expected profit after reinsurance**

The reinsurance premium is given by $1.3E(S_R)$, where:

$$E(S_R) = E(Z)E(N) = \mu E(Z)$$

Now:

$$E(Z) = \int_{1,000}^{\infty} (x - 1,000) \frac{3 \times 1,000^3}{(1,000 + x)^4} dx$$

Setting $u = x - 1,000$:

$$E(Z) = \int_0^{\infty} u \frac{3 \times 1,000^3}{(2,000 + u)^4} du = \left(\frac{1,000}{2,000} \right)^3 \int_0^{\infty} u \frac{3 \times 2,000^3}{(2,000 + u)^4} du$$

Recognising this integral as the mean of the *Pareto*(3, 2000) distribution, we see that:

$$E(Z) = \left(\frac{1}{2} \right)^3 \times \frac{2,000}{3-1} = 125$$

and:

$$E(S_R) = 125\mu$$

So the reinsurance premium is $1.3 \times 125\mu = 162.5\mu$.

Alternatively we could evaluate this integral using the substitution $t = 1000 + x$ or using integration by parts.

The insurer's expected aggregate claim payment is:

$$E(S_I) = E(S) - E(S_R) = 500\mu - 125\mu = 375\mu$$

So the insurer's expected profit after reinsurance is:

$$600\mu - 162.5\mu - 375\mu = 62.5\mu$$

This is the insurer's premium income, minus the premium paid by the insurer to the reinsurer, minus the insurer's expected aggregate claim payment.

The percentage reduction in the expected profit (which was 100μ without reinsurance) is 37.5%.

(iii) **Percentage reduction in standard deviation**

In the absence of reinsurance, the insurer's profit is equal to its premium income minus the aggregate claim amount. Since the premium income is a fixed amount and only the cost of claims is random, the variance of the profit is:

$$\text{var}(S) = \mu E(X^2)$$

We have:

$$E(X) = \frac{\lambda}{\alpha - 1} = 500 \quad \text{and} \quad \text{var}(X) = \frac{\alpha \lambda^2}{(\alpha - 1)^2 (\alpha - 2)} = 750,000$$

So:

$$E(X^2) = \text{var}(X) + [E(X)]^2 = 750,000 + 500^2 = 1,000,000$$

and:

$$\text{var}(S) = 1,000,000\mu$$

Hence the standard deviation of the profit is $1,000\sqrt{\mu}$.

With reinsurance, the insurer's profit is equal to premiums charged less the reinsurance premium less the net claims paid. Since the premiums are fixed amounts, the variance of the insurer's profit is:

$$\text{var}(S_I) = \mu E(Y^2)$$

where:

$$\begin{aligned} E(Y^2) &= \int_0^{1,000} x^2 \frac{3 \times 1,000^3}{(1,000 + x)^4} dx + \int_{1,000}^{\infty} 1,000^2 \frac{3 \times 1,000^3}{(1,000 + x)^4} dx \\ &= 3 \times 1,000^3 \int_0^{1,000} \frac{x^2}{(1,000 + x)^4} dx + 3 \times 1,000^5 \int_{1,000}^{\infty} \frac{1}{(1,000 + x)^4} dx \end{aligned}$$

The second integral is:

$$\int_{1,000}^{\infty} \frac{1}{(1,000 + x)^4} dx = \left[\frac{(1,000 + x)^{-3}}{-3} \right]_{1,000}^{\infty} = \frac{1}{3 \times 2,000^3}$$

For the first integral, we can set $u = 1,000 + x$ to give:

$$\int_0^{1,000} \frac{x^2}{(1,000 + x)^4} dx = \int_{1,000}^{2,000} \frac{(u - 1,000)^2}{u^4} du = \left[-\frac{1}{u} + \frac{1,000}{u^2} - \frac{1,000,000}{3u^3} \right]_{1,000}^{2,000} = \frac{1}{24,000}$$

Alternatively, we could integrate by parts (twice).

So:

$$E(Y^2) = \frac{3 \times 1,000^3}{24,000} + \frac{3 \times 1,000^5}{3 \times 2,000^3} = 250,000 \quad \text{and} \quad \text{var}(S_I) = 250,000\mu$$

Hence the standard deviation of the insurer's profit is now $500\sqrt{\mu}$, which is a reduction of 50%.

The standard deviation is reduced by a greater percentage than the mean. This is very often the case for excess of loss reinsurance.

20.3 We have $X \sim U(0, 200)$ and:

$$Y = \begin{cases} X & \text{if } X \leq M \\ M & \text{if } X > M \end{cases} \quad Z = \begin{cases} 0 & \text{if } X \leq M \\ X - M & \text{if } X > M \end{cases}$$

The expected amount paid by the insurer on any claim is:

$$E(Y) = \int_0^M x \frac{1}{200} dx + \int_M^{200} M \frac{1}{200} dx = 50$$

Solving this:

$$\begin{aligned} \left[\frac{x^2}{400} \right]_0^M + \left[\frac{Mx}{200} \right]_M^{200} &= 50 \\ \Rightarrow \frac{M^2}{400} + M - \frac{M^2}{200} &= 50 \\ \Rightarrow M^2 - 400M + 20,000 &= 0 \\ \Rightarrow M = \frac{-(-400) \pm \sqrt{(-400)^2 - 4 \times 1 \times 20,000}}{2 \times 1} &= 58.579 \text{ or } 341.42 \end{aligned}$$

Since claims are a maximum of 200, M must be 58.579.

The variance of the aggregate annual claims paid by the insurer is:

$$\text{var}(S) = \lambda E(Y^2) = 20E(Y^2)$$

where:

$$\begin{aligned} E(Y^2) &= \int_0^M x^2 \frac{1}{200} dx + \int_M^{200} M^2 \frac{1}{200} dx = \left[\frac{x^3}{600} \right]_0^M + \left[\frac{M^2 x}{200} \right]_M^{200} \\ &= \frac{M^3}{600} + M^2 - \frac{M^3}{200} \\ &= \frac{58.579^3}{600} + 58.579^2 - \frac{58.579^3}{200} \\ &= 2,761.42 \end{aligned}$$

Hence:

$$\text{var}(S) = 20 \times 2,761.42 = 55,228$$

- 20.4 For each age group, the individual claim amounts have a uniform distribution. So the mean and variance of the individual claim distributions are:

$$E(X) = \frac{1}{2}(b + a) = 1,000x$$

$$\text{and: } \text{var}(X) = \frac{1}{12}(b - a)^2 = \frac{20,000^2}{12}$$

Using the individual risk model, the variance of the aggregate claim amount is:

$$\begin{aligned} \text{var}(S) &= \sum_{i=1}^n \{q_i \sigma_i^2 + q_i(1 - q_i) \mu_i^2\} \\ &= 100 \left[(0.0004) \times \frac{20,000^2}{12} + (0.0004)(0.9996) \times 30,000^2 \right] \\ &\quad + 200 \left[(0.001) \times \frac{20,000^2}{12} + (0.001)(0.999) \times 40,000^2 \right] \\ &= 37.32\text{m} + 326.35\text{m} = 363.67\text{m} \end{aligned}$$

Alternatively, we could model the aggregate claim amount from each group as a compound binomial random variable. For example, $N \sim \text{Bin}(100, 0.0004)$ for the 100 lives aged exactly 30. We could then use the formula for $\text{var}(S)$ from the collective risk model.

- 20.5 Under this reinsurance arrangement, we have:

$$S_I = \begin{cases} S & \text{if } S \leq 400 \\ 400 & \text{if } S > 400 \end{cases} \quad S_R = \begin{cases} 0 & \text{if } S \leq 400 \\ S - 400 & \text{if } S > 400 \end{cases}$$

where S is the total monthly claim amount.

Since individual claim amounts must be either 200 or 300, the possible values of S_I are 0, 200, 300, and 400 and:

$$E(S_I) = 0 \times P(S_I = 0) + 200 \times P(S_I = 200) + 300 \times P(S_I = 300) + 400 \times P(S_I = 400)$$

The insurer's aggregate claim amount is 0 if there are no claims. So:

$$P(S_I = 0) = P(N = 0) = \frac{e^{-1.5} \times 1.5^0}{0!} = e^{-1.5}$$

The insurer's aggregate claim amount is 200 if there is one claim and the amount of the claim is 200. So:

$$P(S_I = 200) = P(N = 1, X_1 = 200)$$

Since N and X_1 are independent, we have:

$$P(S_I = 200) = P(N = 1)P(X_1 = 200) = \frac{e^{-1.5} \times 1.5^1}{1!} \times 0.65 = 0.975e^{-1.5}$$

Similarly:

$$P(S_I = 300) = P(N = 1, X_1 = 300) = \frac{e^{-1.5} \times 1.5^1}{1!} \times 0.35 = 0.525e^{-1.5}$$

Finally, the insurer's aggregate claim amount is 400 if the total claim amount is 400 or more. This probability can be calculated by subtraction as follows:

$$\begin{aligned} P(S_I = 400) &= 1 - P(S_I = 0) - P(S_I = 200) - P(S_I = 300) \\ &= 1 - e^{-1.5} - 0.975e^{-1.5} - 0.525e^{-1.5} \\ &= 1 - 2.5e^{-1.5} \end{aligned}$$

So:

$$E(S_I) = 0 \times e^{-1.5} + 200 \times 0.975e^{-1.5} + 300 \times 0.525e^{-1.5} + 400(1 - 2.5e^{-1.5}) = 255.52$$

We can now calculate $E(S_R)$ using the result:

$$E(S_R) = E(S) - E(S_I)$$

We have:

$$E(S) = \lambda E(X) = 1.5[200 \times 0.65 + 300 \times 0.35] = 1.5 \times 235 = 352.50$$

Hence:

$$E(S_R) = 352.50 - 255.52 = 96.98$$

20.6 This is part of Subject 106, April 2003, Question 9.

(i) **Mean and variance of T**

Let $T = Y_1 + Y_2 + \dots + Y_{500}$, where Y_i is the total claim on the i th policy. Then:

$$E[T] = E[Y_1] + E[Y_2] + \dots + E[Y_{500}]$$

$$\text{var}[T] = \text{var}[Y_1] + \text{var}[Y_2] + \dots + \text{var}[Y_{500}]$$

Since $E[Y_i] = q_i \mu$ and $\text{var}[Y_i] = q_i \sigma^2 + q_i(1 - q_i)\mu^2$, we have:

$$E[T] = \mu \sum_{i=1}^{500} q_i \quad [2]$$

$$\text{var}[T] = \sigma^2 \sum_{i=1}^{500} q_i + \mu^2 \sum_{i=1}^{500} q_i(1 - q_i) \quad [2]$$

since the risks are independent.

(ii) **Mean and variance of $\tilde{T}$**

Let C be the amount claimed in one year on a single risk. Then, according to the approximation:

$$C = X_1 + X_2 + \cdots + X_N$$

where $N \sim \text{Poi}(q_i)$, $E[X] = \mu$ and $\text{var}[X] = \sigma^2$.

Also:

$$\tilde{T} = C_1 + C_2 + \cdots + C_{500}$$

where C_i is the total amount claimed on the i th risk.

Using the formulae for the mean and variance of compound Poisson random variable:

$$E[C_i] = \mu q_i \quad \text{var}[C_i] = q_i(\mu^2 + \sigma^2) \quad [2]$$

Since $\tilde{T}$ is the sum of claims for the whole portfolio, we have:

$$E[\tilde{T}] = \mu \sum_{i=1}^{500} q_i \quad \text{var}[\tilde{T}] = (\sigma^2 + \mu^2) \sum_{i=1}^{500} q_i \quad [2]$$

The mean is the same but the variance is larger than that obtained in part (i).

(iii) **MGF**

By definition we have:

$$M_T(t) = E[e^{tT}] = E[e^{t(Y_1 + Y_2 + \cdots + Y_{500})}] = M_{Y_1}(t) M_{Y_2}(t) \cdots M_{Y_{500}}(t) \quad [1/2]$$

From the information given in the question, Y_i is either 0 with probability 0.98 or μ with probability 0.02. We can therefore work out the moment generating function of Y_i :

$$M_Y(t) = E[e^{tY}] = e^{t \times 0} \times 0.98 + e^{t\mu} \times 0.02 = 0.98 + 0.02e^{t\mu} \quad [1/2]$$

Substituting this into the expression for the moment generating function for T , we get:

$$M_T(t) = (0.98 + 0.02e^{t\mu})^{500} \quad [1/2]$$

This is of the form of the moment generating function for a compound binomial distribution with parameters 500 and 0.02, and claim size distribution that is constant. [1/2]

(iv) **Compound Poisson**

By definition we have:

$$M_{\tilde{T}}(t) = E[e^{t\tilde{T}}] = E[e^{t(C_1 + C_2 + \cdots + C_{500})}] = M_{C_1}(t) M_{C_2}(t) \cdots M_{C_{500}}(t)$$

From the information given in the question, since C_i has a compound Poisson distribution it has moment generating function:

$$M_{C_i}(t) = \exp[q_i(M_X(t) - 1)] = \exp[0.02(M_X(t) - 1)] \quad [\frac{1}{2}]$$

The random variable X takes the value μ with probability 1, so:

$$M_X(t) = e^{t\mu}$$

and:

$$M_{C_i}(t) = \exp[0.02(e^{t\mu} - 1)] \quad [\frac{1}{2}]$$

Substituting this into the expression for the moment generating function for $\tilde{T}$, we get:

$$\begin{aligned} M_{\tilde{T}}(t) &= \exp[0.02(e^{t\mu} - 1)] \exp[0.02(e^{t\mu} - 1)] \dots \exp[0.02(e^{t\mu} - 1)] \\ &= \exp[10(e^{t\mu} - 1)] \end{aligned} \quad [\frac{1}{2}]$$

This is of the form of the moment generating function for a compound Poisson distribution with parameter 10, and claim size distribution that is constant. [$\frac{1}{2}$]

20.7 (i) **Total number of claimants**

The aggregate amount paid out by the company is 1,000S, where:

$S = Z_1 + \dots + Z_N$ is the total number of employees claiming benefit in a year

$$Z = \begin{cases} Y & Y < 3 \\ 3 & Y \geq 3 \end{cases}$$

$$Y = X + 1$$

and:

$$X \sim \text{Poisson}(2.2)$$

Now:

$$P(Y = 1) = P(X = 0) = e^{-2.2} \quad [\frac{1}{2}]$$

$$P(Y = 2) = P(X = 1) = 2.2e^{-2.2} \quad [\frac{1}{2}]$$

$$P(Y \geq 3) = 1 - e^{-2.2} - 2.2e^{-2.2} = 1 - 3.2e^{-2.2} \quad [\frac{1}{2}]$$

So:

$$\begin{aligned} E(Z) &= 1P(Y=1) + 2P(Y=2) + 3P(Y \geq 3) \\ &= e^{-2.2} + 4.4e^{-2.2} + 3(1 - 3.2e^{-2.2}) = 3 - 4.2e^{-2.2} = 2.53463 \end{aligned} \quad [1]$$

and:

$$\begin{aligned} E(Z^2) &= 1^2P(Y=1) + 2^2P(Y=2) + 3^2P(Y \geq 3) \\ &= e^{-2.2} + 8.8e^{-2.2} + 9(1 - 3.2e^{-2.2}) = 9 - 19e^{-2.2} = 6.89474 \end{aligned} \quad [1]$$

Hence the variance of Z is:

$$\text{var}(Z) = E(Z^2) - (E(Z))^2 = 6.89474 - 2.53463^2 = 0.47041 \quad [\frac{1}{2}]$$

To find $E(N)$ and $\text{var}(N)$, we use the fact that N has a Type 2 negative binomial distribution with parameters $p=0.8$, $q=0.2$ and $k=1$. Using the formulae for the moments given on page 9 the *Tables*, we have:

$$E(N) = \frac{kq}{p} = \frac{0.2}{0.8} = 0.25 \quad [\frac{1}{2}]$$

$$\text{and: } \text{var}(N) = \frac{kq}{p^2} = \frac{0.2}{0.8^2} = 0.3125 \quad [\frac{1}{2}]$$

Alternatively, we could derive the moment generating function of N , and then use MGF formulae to derive the mean and variance of N .

The mean and variance of S are:

$$E(S) = E(Z)E(N) = 2.53463 \times 0.25 = 0.63366 \quad [1]$$

and:

$$\text{var}(S) = (E(Z))^2 \text{var}(N) + \text{var}(Z)E(N) = 2.53463^2 \times 0.3125 + 0.47041 \times 0.25 = 2.12521 \quad [1]$$

(ii) **Mean and variance of the aggregate amount**

So the mean and variance of 1,000S are:

$$E(1,000S) = 1,000E(S) = 634 \quad [\frac{1}{2}]$$

and:

$$\text{var}(1,000S) = 1,000^2 \text{var}(S) = 2,125,000 \quad [\frac{1}{2}]$$

Alternatively, we could define $S = Z_1 + Z_2 + \dots + Z_N$, where:

$$Z = \begin{cases} 1,000Y & \text{if } Y \leq 3 \\ 3,000 & \text{if } Y > 3 \end{cases}$$

This would give us the aggregate claim amount directly.

20.8 (i) **Mean, variance and coefficient of skewness**

We have $N_1 \sim \text{Poisson}(400 \times \frac{1}{20}) \equiv \text{Poisson}(20)$, $X_1 = \text{£}3,000$, $N_2 \sim \text{Poisson}(250 \times \frac{1}{10}) \equiv \text{Poisson}(25)$ and:

$$X_2 = \begin{cases} \text{£}2,000 & \text{with probability } 0.4 \\ \text{£}3,000 & \text{with probability } 0.6 \end{cases}$$

Working in £000s, we find that:

$$\begin{aligned} E(X_2) &= (2 \times 0.4) + (3 \times 0.6) = 2.6 \\ E(X_2^2) &= (2^2 \times 0.4) + (3^2 \times 0.6) = 7 \\ E(X_2^3) &= (2^3 \times 0.4) + (3^3 \times 0.6) = 19.4 \end{aligned} \quad [1]$$

Let S_i denote the annual aggregate claims from category i . Using the assumption that the policies are independent and the result that, for a compound Poisson random variable T , the k th central moment of T is given by $\lambda E(X^k)$, we obtain:

$$E(S) = E(S_1) + E(S_2) = (20 \times 3) + (25 \times 2.6) = 125 = \text{£}125,000 \quad [1]$$

$$\begin{aligned} \text{var}(S) &= \text{var}(S_1) + \text{var}(S_2) \\ &= \lambda_1 E(X_1^2) + \lambda_2 E(X_2^2) \\ &= (20 \times 9) + (25 \times 7) \\ &= 355 \\ &= \text{£}^2 355,000,000 \end{aligned} \quad [1]$$

$$\begin{aligned} \text{skew}(S) &= \text{skew}(S_1) + \text{skew}(S_2) \\ &= \lambda_1 E(X_1^3) + \lambda_2 E(X_2^3) \\ &= (20 \times 27) + (25 \times 19.4) \\ &= 1,025 \\ &= \text{£}^3 (1,025 \times 10^9) \end{aligned}$$

So the coefficient of skewness is:

$$\frac{\text{skew}(S)}{(\text{var}(S))^{3/2}} = \frac{1,025}{355^{3/2}} = 0.15324 \quad [1]$$

(ii) **Calculate Y using a normal approximation**

Assuming that $S \sim N(125, 355)$, we have:

$$\begin{aligned} P(S > Y) = 0.1 &\Rightarrow P\left(N(0,1) > \frac{Y-125}{\sqrt{355}}\right) = 0.1 \\ &\Rightarrow \frac{Y-125}{\sqrt{355}} = 1.2816 \quad \text{from page 162 Tables} \\ &\Rightarrow Y = 149.147 \end{aligned}$$

So S exceeds £149,000 with a probability of approximately 0.1. [3]

(iii) **Probability that reinsurer is involved**

The expected value and variance of S are now the same as those of S_1 . Working in £000s and assuming that $S \sim N(60, 180)$, we obtain:

$$P(S > 100) = P\left(N(0,1) > \frac{100-60}{\sqrt{180}}\right) \approx 1 - \Phi(2.98) = 1 - 0.9986 = 0.0014 \quad [3]$$

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